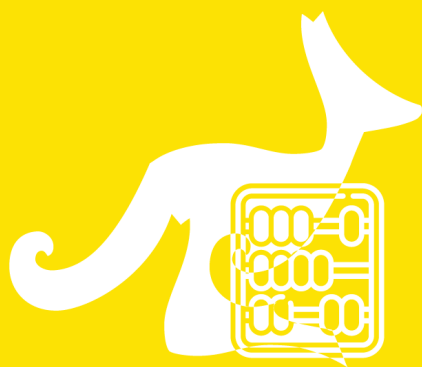




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a

2B pencil.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

Good Luck!

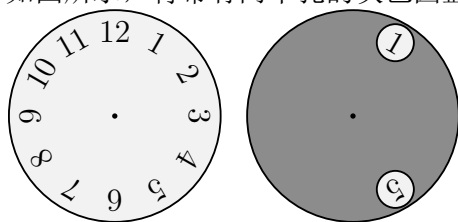
预祝考试顺利

**Grade 9
&
Grade 10**

3 points

1. A grey circle with two holes is put on top of a clock-face, as shown.

如图所示，将带有两个孔的灰色圆盘放在钟面上。



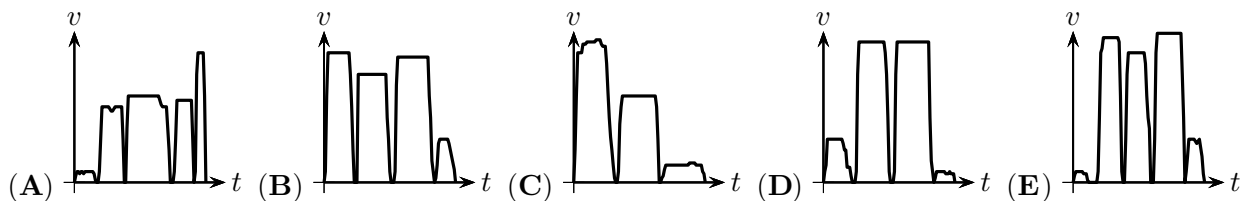
The grey circle is turned around the centre such that the number 10 appears in one hole. Which numbers is it possible to see in the other hole?

灰色圆盘绕其中心旋转，一个孔中出现数字 10。另一个孔中可能看到哪些数字？

- (A) 2 and 6 2 和 6 (B) 3 and 7 3 和 7 (C) 3 and 6 3 和 6
(D) 1 and 9 1 和 9 (E) 2 and 7 2 和 7

2. Maria had to run to catch the subway, got off two stops later and then walked to school. Which of the following speed-time graphs would best represent her journey?

Maria 必须跑着去乘坐地铁，两站后下车，然后步行去学校。以下哪个速度和时间的关系图像最符合 Maria 的行程？



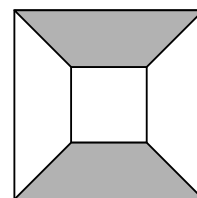
3. The positive integers m and n are both odd. Which of the following integers is also odd?

正整数 m 和 n 都是奇数。下面哪个整数也是奇数？

- (A) $m(n+1)$ (B) $(m+1) \cdot (n+1)$ (C) $m+n+2$
(D) $m \cdot n+2$ (E) $m+n$

4. A large square of side-length 10 cm contains a smaller square of side-length 4 cm, as shown in the diagram. The corresponding sides of the two squares are parallel. What percentage of the large square is shaded?

边长为 10 cm 的大正方形内部有一个边长为 4 cm 的小正方形，如图所示。两个正方形相对应的边是平行的。阴影部分占大正方形的百分比是多少？



- (A) 25% (B) 30% (C) 40%
(D) 42% (E) 45%

5. Today is Thursday. What day will it be in 2023 days' time?

今天是星期四。2023 天后会是星期几?

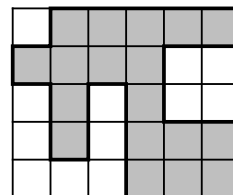
- (A) Tuesday 星期二 (B) Wednesday 星期三 (C) Thursday 星期四
(D) Friday 星期五 (E) Saturday 星期六

6. The large rectangle in the diagram is divided into 30 equal squares, as shown.

The perimeter of the shaded region is 240 cm. What is the area of the rectangle?

如图所示, 大长方形被分成 30 个相等的正方形。阴影部分的周长是 240 cm。

这个大长方形的面积是多少?



- (A) 480 cm² (B) 750 cm² (C) 1080 cm²
(D) 1920 cm² (E) 2430 cm²

7. The ages of a family of five add to 80. The two youngest are 6 and 8. What was the sum of the ages of the family seven years ago?

一个五口之家中所有人的年龄之和为 80 岁, 最小的两位家庭成员分别是 6 岁和 8 岁。七年前这家人的年龄之和是多少?

- (A) 35 (B) 36 (C) 45 (D) 46 (E) 66

8. A wooden fence consists of a series of vertical planks, each joined to the next post by four horizontal planks. The first and last plank in the fence are vertical. Which of the following could be the total number of planks in the fence?

一个木栅栏由许多垂直木板组成, 每块木板之间连接有四块水平木板。栅栏的第一块和最后一块木板都是垂直的。以下哪项可能是组成栅栏的木板总数?

- (A) 95 (B) 96 (C) 97 (D) 98 (E) 99

9. The letters a and b are to be replaced by positive integers so that the equation is correct.

$$\frac{a}{5} = \frac{7}{b}$$

In how many different ways can this be done?

有多少种不同的方式用正整数代替字母 a 和 b , 使得等式成立?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

10. After having played 200 games of chess, my winning rate is exactly 49%. What is the smallest number of extra games I need to play to increase my winning rate to exactly 50%?

下了 200 盘象棋后, 我的获胜率恰好是 49%。要将我的获胜率提高到恰好是 50%, 我最少还需要下多少盘棋?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

4 points

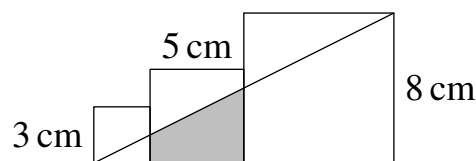
11. Jenni is trying to save water. She reduced the time she spent in her shower by a quarter. She also lowered the water pressure of her shower to reduce the rate the water comes out of the shower head by a quarter. By what fraction did Jenni reduce the total amount of water she uses for a shower?

Jenni 正在努力节约用水。她将淋浴的时间减少了四分之一，同时还降低了淋浴器的水压，使得淋浴头出水的速度降低了四分之一。Jenni 淋浴的总用水量一共减少了多少？

- (A) $\frac{1}{4}$ (B) $\frac{3}{8}$ (C) $\frac{5}{8}$ (D) $\frac{5}{12}$ (E) $\frac{7}{16}$

12. The diagram shows three squares of side-length 3 cm, 5 cm and 8 cm. What is the area, in cm^2 , of the shaded trapezium?

该图显示了边长分别为 3 厘米、5 厘米和 8 厘米的三个正方形，阴影梯形的面积是多少平方厘米？



- (A) 13 (B) $\frac{55}{4}$ (C) $\frac{61}{4}$ (D) $\frac{65}{4}$ (E) $\frac{69}{4}$

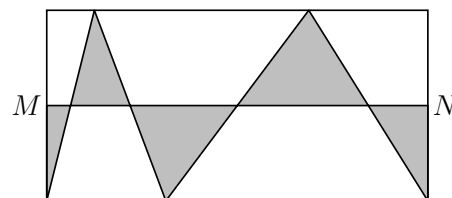
13. A wire of length 95 m is cut into three pieces such that the length of each piece is 50% more than the previous piece. What is the length of the largest piece?

长度为 95 m 的金属丝被切成三段，且每段的长度都比前一段长 50%。最长一段的长度是多少？

- (A) 36 m (B) 42 m (C) 45 m (D) 46 m (E) 48 m

14. Points M and N are the midpoints of two sides of the rectangle. What fraction of the area of the rectangle is shaded?

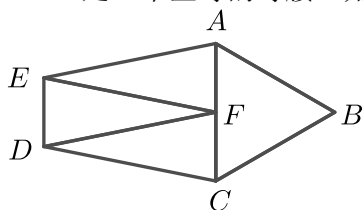
点 M 和 N 是长方形两条边的中点。长方形中阴影部分占总面积的几分之几？



- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

15. Pentagon $ABCDE$ is divided into four triangles with equal perimeter. Triangle ABC is equilateral and AEF , DFE and CDF are three identical isosceles triangles. What is the ratio of the perimeter of the pentagon $ABCDE$ to the perimeter of triangle ABC ?

五边形 $ABCDE$ 被分为四个周长相等的三角形。三角形 ABC 是等边三角形， AEF 、 DFE 和 CDF 是三个全等的等腰三角形。五边形 $ABCDE$ 与三角形 ABC 的周长之比是多少？



- (A) $\frac{2}{1}$ (B) $\frac{3}{2}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$ (E) $\frac{5}{2}$

16. On the table there is a tower made of blocks numbered from 1 to 90. Bob takes blocks from the top of the tower, three at a time, to build a new tower, as shown. When he has finished building the new tower, how many blocks will be between the blocks numbered 39 and 40?

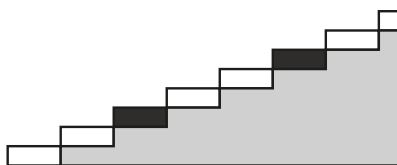
90	3
89	2
88	1
⋮	⋮
4	85
3	90
2	89
1	88

桌子上有一个由编号 1 到 90 的积木组成的塔。如图所示, Bob 从塔顶一次取三块积木, 建造一个新塔。当建造完成后, 编号为 39 和 40 的积木之间有多少块积木?

- (A) 0 (B) 1 (C) 2
(D) 3 (E) 4

17. Every third step of a staircase with 2023 steps is coloured black. The first seven steps are shown in the diagram. Anita walks up the steps one at a time, starting with either her right or left foot, alternating each step.

一个楼梯有 2023 级台阶, 每过两个台阶就有一个涂黑的台阶, 图中显示了前七级台阶。Anita 左右脚交替, 一次走上一级台阶。



What is the smallest number of black steps she will step on with her right foot?

她的右脚最少踩过多少个黑色台阶?

- (A) 0 (B) 333 (C) 336 (D) 337 (E) 674

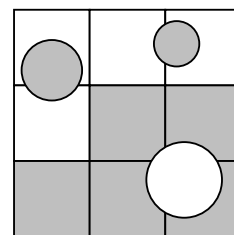
18. We call a two-digit number **power-less** if none of its digits can be written as an integer to a power greater than 1. For example, 53 is power-less, but 54 is **NOT** power-less since $4 = 2^2$. Which of the following is a common divisor of the smallest and the largest power-less numbers?

一个两位数, 如果它的各个数位上的数字都不能写成一个指数大于 1 的整数, 那么我们就称这个两位数为 **无幂数**。例如, 53 是无幂数, 但 54 不是无幂数, 因为 $4 = 2^2$ 。以下哪项是最小无幂数和最大无幂数的公约数?

- (A) 3 (B) 5 (C) 7 (D) 11 (E) 13

19. A square of side 30 cm is divided into nine identical smaller squares. The large square contains three circles with radii 5 cm (bottom right), 4 cm (top left) and 3 cm (top right), as shown. What is the area of the shaded part?

一个边长为 30 cm 的正方形被分成 9 个相同的小正方形。如图所示, 大正方形包含三个半径分别为 5 cm (右下角)、4 cm (左上角) 和 3 cm (右上角) 的圆。阴影部分的面积是多少?



- (A) 400 cm^2 (B) 500 cm^2 (C) $(400 + 50\pi) \text{ cm}^2$
(D) $(500 - 25\pi) \text{ cm}^2$ (E) $(500 + 25\pi) \text{ cm}^2$

20. Tim calculates the mean of five different prime numbers. His answer is an integer. What is the smallest possible integer he could have obtained?

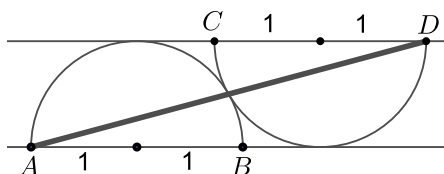
Tim 计算 5 个不同质数的平均数，得到的答案是一个整数。这个整数的最小可能值是多少？

- (A) 2 (B) 5 (C) 6 (D) 12 (E) 30

5 points

21. The diagram shows two touching semicircles of radius 1 and parallel diameters AB and CD . What is the square of the distance AD ?

如图所示，两个半径为 1 的半圆相切，直径 AB 和 CD 互相平行。 AD 之间距离的平方是多少？



- (A) 16 (B) $8 + 4\sqrt{3}$ (C) 12 (D) 9 (E) $5 + 2\sqrt{3}$

22. When it is given a list of four numbers, the Kangaroo Machine continues the list by typing the smallest non-negative integer that is different to each of the four preceding terms and then repeats this process over and over again. Jacob types in the numbers

2, 0, 2, 3,

into the machine. What number will be the 2023rd in the list?

将 4 个数字构成的数列输入袋鼠机器后，机器会输出一个与前面四项都不相同的最小非负整数来继续数列，然后不断重复这个过程。Jacob 把数列

2, 0, 2, 3,

输入袋鼠机器，请问数列中第 2023 项是多少？

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

23. A rectangle with vertices $(0,0)$, $(100,0)$, $(100,50)$ and $(0,50)$ has a circle with centre $(75,30)$ and radius 10 cut out of it. What is the gradient of the line through $(75,30)$ that divides the remaining area of the rectangle into two equal parts?

从顶点坐标为 $(0,0)$, $(100,0)$, $(100,50)$ 和 $(0,50)$ 的长方形中挖去一个圆心坐标为 $(75,30)$ ，半径为 10 的圆。过点 $(75,30)$ 且将长方形剩余面积分成两个相等部分的直线斜率是多少？

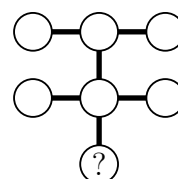
- (A) $\frac{1}{5}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{5}$ (E) $\frac{2}{3}$

24. When Metin's phone is fully charged, it runs out in 32 hours if he only uses it for phone-calls, in 20 hours if he only uses it for the internet, and in 80 hours if he does not use it at all. Metin gets on a train with his phone half-charged. While on the train, the time he is on the internet, the time he is making phone-calls and the time he is not using it are all the same. His phone runs out of charge just as the train reaches his destination. How many hours did the train journey take?

Metin 把他的手机充满电后，如果他只用于打电话，32 小时内会耗尽电量；如果只用于上网，20 小时内会耗尽电量；如果不使用手机，80 小时内会耗尽电量。Metin 上火车时，手机电量还有一半。在火车上，他使用手机上网、打电话和不使用手机的时长都相同。火车刚到目的地，他的手机电量就全部耗尽了。请问他一共坐了多少个小时的火车？

- (A) 10 (B) 12 (C) 15 (D) 16 (E) 18

25. Seven different single-digit numbers are written in the circles of the diagram shown with one number in each circle. The product of the three numbers in each of the three lines of three numbers is the same. Which number is written in the circle containing the question mark?

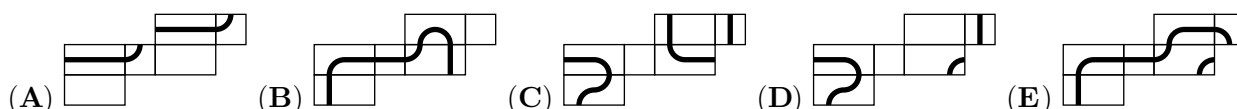


将 7 个不同的一位数填入右图圆圈中，每个圆圈中填入一个数字。位于三条直线上的三个数字之积是相同的。问号圆圈中填写的是哪个数字？

- (A) 2 (B) 3 (C) 4 (D) 6 (E) 8

26. Leon has drawn a closed path on a cuboid and then unfolded it to give a net. Which of the nets shown could not be the net of Leon's cuboid?

Leon 在一个长方体上画了一条封闭路径，然后将其展开得到长方体的展开图。以下哪个不可能是 Leon 所画长方体的展开图？



27. How many three-digit positive integers x are there, such that subtracting the sum of digits of x from x gives a three-digit number whose digits are all the same?

有多少个三位正整数 x ，满足 x 减去 x 各个数位上的数字之和后，得到一个各个数位上的数字都相同的三位数？

- (A) 1 (B) 2 (C) 3 (D) 20 (E) 30

28. In how many different ways can the word **BANANA** be read from the following table by moving from one cell to another cell with which it shares an edge? Cells may be visited more than once.

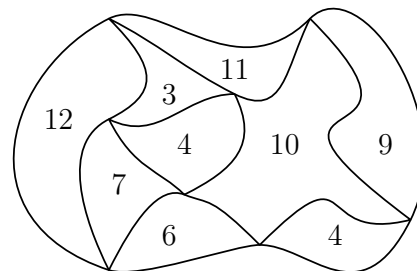
B	A	N
A	N	A
N	A	N

可以从表格中的一个单元格读到与其共用一条边的另一个单元格，一共有多少种不同的方式读取单词 **BANANA**？（同一个单元格可以多次读取）

- (A) 14 (B) 28 (C) 56 (D) 84 (E) other value 以上都不是

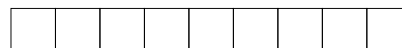
29. The diagram shows a map of a park. The park is divided into regions. The number inside each region gives its perimeter, in km. What is the outer perimeter of the park?

右图是一张公园地图。公园分为多个区域，每个区域内的数字表示这个区域的周长，单位为千米。公园的外围周长是多少千米？



- (A) 22 km 22 千米 (B) 26 km 26 千米
(C) 28 km 28 千米 (D) 32 km 32 千米
(E) none of the previous 以上都不正确

30. Pia wants to write the integers 1 to 9 in the nine boxes shown so that the integers in any three adjacent boxes add to a multiple of 3. In how many ways can she do this?



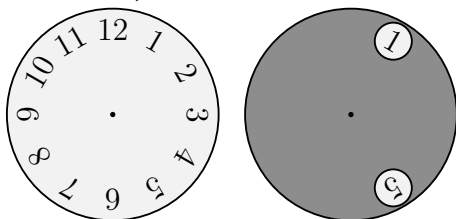
Pia 可以有多少种方式将整数 1 到 9 填入右侧九个方框中，使得任意三个相邻方框中的整数之和是 3 的倍数？

- (A) 6^4 (B) 6^3 (C) 2^9
(D) $6 \times 5 \times 4 \times 3 \times 2 \times 1$ (E) $9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

Junior

3 points

1 (Denmark). A grey circle with two holes is put on top of a clock-face, as shown.

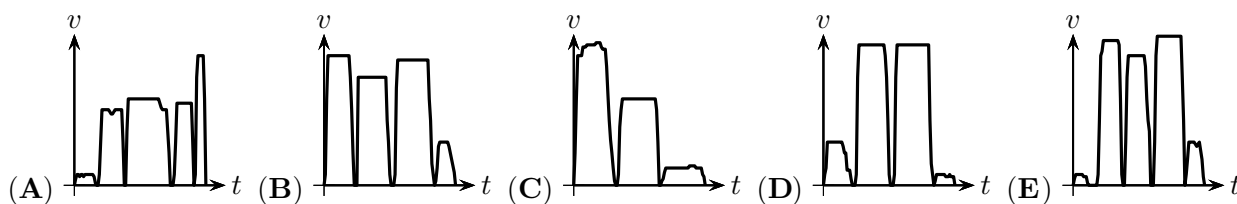


The grey circle is turned around the centre such that the number 10 appears in one hole. Which numbers is it possible to see in the other hole?

- (A) 2 and 6 (B) 3 and 7 (C) 3 and 6 (D) 1 and 9 (E) 2 and 7

SOLUTION: The picture shows that one hole will show a time 4 hours after the other. This is only true for A.

2 (Germany). Maria had to run to catch the subway, got off two stops later and then walked to school. Which of the following speed-time graphs would best represent her journey?



SOLUTION: Compare the speeds of the three versions of moving and consider the breaks in between.

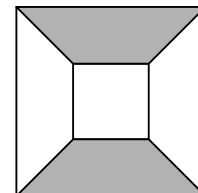
3 (Sweden). The positive integers m and n are both odd. Which of the following integers is also odd?

- (A) $m(n+1)$ (B) $(m+1) \cdot (n+1)$ (C) $m+n+2$
(D) $m \cdot n+2$ (E) $m+n$

SOLUTION: When m and n are both odd then $m+1$, $n+1$ and $n+m$ are all even. That's why the integers in (A), (B), (C) and (E) are all even. Only (D) is still odd.

4 (Slovakia). A large square of side-length 10 cm contains a smaller square of side-length 4 cm, as shown in the diagram. The corresponding sides of the two squares are parallel. What percentage of the large square is shaded?

- (A) 25% (B) 30% (C) 40%
(D) 42% (E) 45%



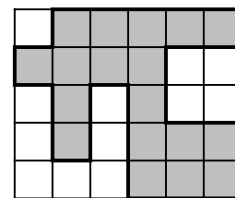
SOLUTION: Because the big square is of 10 cm and the smaller is of 4 cm the sum of the heights of the two trapezoids is 6 cm. Then we got for the area of both trapezoids $\frac{10 \text{ cm} + 4 \text{ cm}}{2} \cdot 6 \text{ cm} = 42 \text{ cm}^2$.

5 (Hungary). Today is Thursday. What day will it be in 2023 days' time?

- (A) Tuesday (B) Wednesday (C) Thursday (D) Friday (E) Saturday

SOLUTION: Because 2023 is divisible by 7, this day again will be a Thursday.

6 (Mexico). The large rectangle in the diagram is divided into 30 equal squares, as shown. The perimeter of the shaded region is 240 cm. What is the area of the rectangle?



- (A) 480 cm^2 (B) 750 cm^2 (C) 1080 cm^2
 (D) 1920 cm^2 (E) 2430 cm^2

SOLUTION: The shaded region has 4 more vertical sides and 4 more horizontal sides of small squares than the rectangle, this is $2 \times (5 + 6) + 8 = 30$. Hence the length of each side of a small square is $240 \text{ cm} / 30 = 8 \text{ cm}$ and the rectangle has area $30 \times 8 \text{ cm} \times 8 \text{ cm} = 1920 \text{ cm}^2$.

7 (United Kingdom). The ages of a family of five add to 80. The two youngest are 6 and 8. What was the sum of the ages of the family seven years ago?

- (A) 35 (B) 36 (C) 45 (D) 46 (E) 66

SOLUTION: The sum of the four eldest members of the family is 74. Seven years ago, the sum of their ages was $74 - 28 = 46$. Seven years ago, the youngest child had not been born and so did not contribute to the sum of their ages.

8 (Italy). A wooden fence consists of a series of vertical planks, each joined to the next post by four horizontal planks. The first and last plank in the fence are vertical. Which of the following could be the total number of planks in the fence?

- (A) 95 (B) 96 (C) 97 (D) 98 (E) 99

SOLUTION: The first field of this fence consists of 6 planks (2 vertical and 4 horizontal planks), every next field consists of 5 planks (another vertical and 4 horizontal planks). So the number must be some number of the form $5 \cdot x + 1$ ($x \geq 1$).

9 (Germany). The letters a and b are to be replaced by positive integers so that the equation is correct. In how many different ways can this be done? $\frac{a}{5} = \frac{7}{b}$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

SOLUTION: $a \cdot b = 5 \cdot 7$ is equivalent to the given equation. Since the prime decomposition of 35 is $5 \cdot 7$ all ordered pairs of positive integers that fulfill this equation are (1, 35), (35, 1), (5, 7) and (7, 5).

10 (Paraguay). After having played 200 games of chess, my winning rate is exactly 49%. What is the smallest number of extra games I need to play to increase my winning rate to exactly 50%?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

SOLUTION: Winning 49% of 200 games means winning 98 games and not winning 102. For me to have equal number of wins and NOT wins I need to win $102 - 98 = 4$ games to have 102 wins meaning 50% rate of success.

4 points

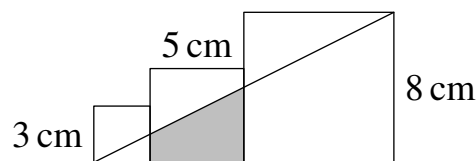
11 (Germany). Jenni is trying to save water. She reduced the time she spent in her shower by a quarter. She also lowered the water pressure of her shower to reduce the rate the water comes out of the shower head by a quarter. By what fraction did Jenni reduce the total amount of water she uses for a shower?

- (A) by $\frac{1}{4}$ (B) by $\frac{3}{8}$ (C) by $\frac{5}{8}$ (D) by $\frac{5}{12}$ (E) by $\frac{7}{16}$

SOLUTION: In total Jenni uses $\frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}$ of the water volume that she used before. So she reduced the total amount of water by $\frac{7}{16}$.

12 (United Kingdom). The diagram shows three squares of side-length 3 cm, 5 cm and 8 cm. What is the area, in cm^2 , of the shaded trapezium?

- (A) 13 (B) $\frac{55}{4}$ (C) $\frac{61}{4}$ (D) $\frac{65}{4}$ (E) $\frac{69}{4}$



SOLUTION: Let the lengths of the vertical sides of the trapezium be p and q . Therefore, using similar triangles, $\frac{p}{3} = \frac{q}{3+5} = \frac{8}{3+5+8}$. Hence $p = \frac{3}{2}$ and $q = 4$. Therefore the area of the trapezium is $\frac{1}{2} \times \left[\frac{3}{2} + 4 \right] \times 5 = \frac{55}{4}$.

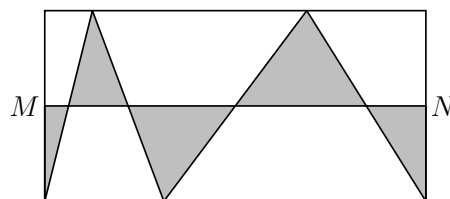
13 (Catalonia). A wire of length 95 m is cut into three pieces such that the length of each piece is 50% more than the previous piece. What is the length of the largest piece?

- (A) 36 m (B) 42 m (C) 45 m (D) 46 m (E) 48 m

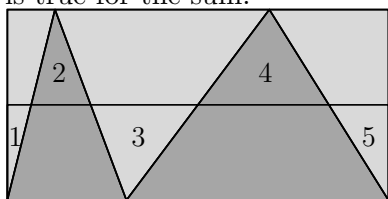
SOLUTION: It may be convenient to go backwards. If a value v is the previous increased by half, then the previous is $2v/3$. So if v is the requested length, the three lengths are $4v/9 + 2v/3 + v = 95$ m and hence $v = 45$ m.

14 (Paraguay). Points M and N are the midpoints of two sides of the rectangle. What fraction of the area of the rectangle is shaded?

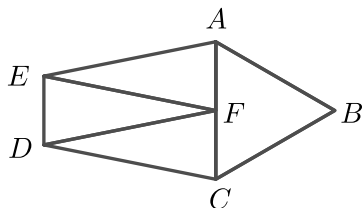
- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$



SOLUTION: Because their height and base are half the size of the larger triangles, each of the five shaded small triangles has an area equal to $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ of the corresponding large triangle. The same is true for the sum.

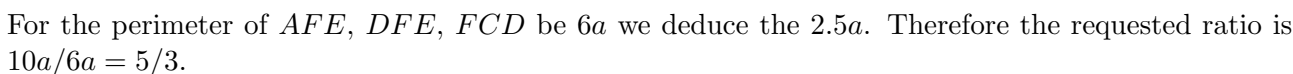


15 (Catalonia). Pentagon $ABCDE$ is divided into four triangles with equal perimeter. Triangle ABC is equilateral and AEF , DFE and CDF are three identical isosceles triangles. What is the ratio of the perimeter of the pentagon $ABCDE$ to the perimeter of triangle ABC ?



- (A) $2/1$ (B) $3/2$ (C) $4/3$ (D) $5/3$ (E) $5/2$

SOLUTION: We call $2a$ the side of equilateral triangle. The figure shows the length of all segments.



90	3
89	2
88	1
⋮	⋮
4	85
3	90
2	89
1	88

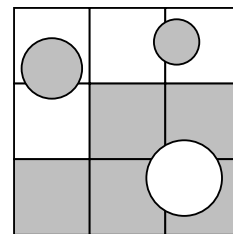
- SOLUTION:** In the new tower, under each block numbered by an integer divisible by 3, there will be successively blocks numbered by integers lower by 1 and by 2. The number 39 is divisible by 3, so there will be blocks numbered 38 and 37 underneath it. Under the block numbered 37 there will be the block numbered by the next integer after 39 that is divisible by 3, i.e. 42, and under it blocks numbered 41 and 40. It follows that between blocks numbered 39 and 40 there will be 4 other blocks (numbered 38, 37, 42 and 41).

(A) 0 (B) 333 (C) 336 (D) 337 (E) 674

18 (United Arab Emirates). We call a two-digit number **power-less** if none of its digits can be written as an integer to a power greater than 1. For example, 53 is power-less, but 54 is **NOT** power-less since $4 = 2^2$. Which of the following is a common divisor of the smallest and the largest power-less numbers?

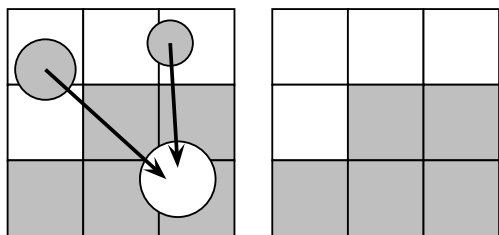
- SOLUTION: The digits of a **power-less** number can be among 2, 3, 5, 6, and 7 only since both 0 and 1 can be written as any power of themselves and $9 = 3^2$, $8 = 2^3$, $4 = 2^2$. The smallest two-digit number is 22 and the largest one is 77 and it is easy to see ($22 = 2 \cdot 11$, $77 = 7 \cdot 11$) that they have only one common divisor greater than 1, that is 11.

19 (Greece). A square of side 30 cm is divided into nine identical smaller squares. The large square contains three circles with radii 5 cm (bottom right), 4 cm (top left) and 3 cm (top right), as shown. What is the area of the shaded part?



- (A) 400 cm^2 (B) 500 cm^2 (C) $(400 + 50\pi) \text{ cm}^2$
 (D) $(500 - 25\pi) \text{ cm}^2$ (E) $(500 + 25\pi) \text{ cm}^2$

SOLUTION: The idea is NOT to start adding and subtracting the areas of individual parts but to compare them. Note that the sum of the areas of the two smaller grey circles is the same as that of the big white circle since $3^2\pi + 4^2\pi = 5^2\pi$. So exchanging the two grey circles with the white one, the region becomes a simple polygon whose area is $\frac{5}{9} \times (30 \text{ cm})^2 = 500 \text{ cm}^2$.



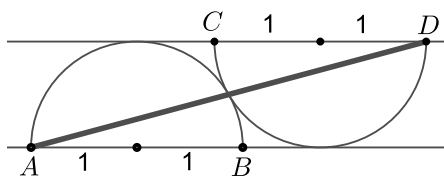
20 (Paraguay). Tim calculates the mean of five different prime numbers. His answer is an integer. What is the smallest possible integer he could have obtained?

- (A) 2 (B) 5 (C) 6 (D) 12 (E) 30

SOLUTION: The sum of 5 primes must be divisible by 5. The sum of the first 5 primes $2+3+5+7+11 = 28$ does not work. Increasing 11 to 13 we get a solution for the average of 6: $2+3+5+7+13 = 30$.

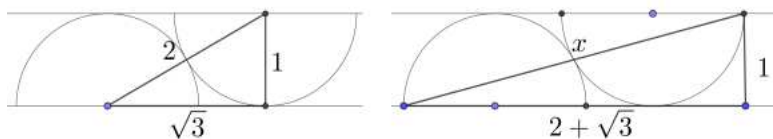
5 points

21 (Catalonia). The diagram shows two touching semicircles of radius 1 and parallel diameters AB and CD . What is the square of the distance AD ?



- (A) 16 (B) $8 + 4\sqrt{3}$ (C) 12 (D) 9 (E) $5 + 2\sqrt{3}$

SOLUTION: Two applications of the Pythagorean theorem.



$$(2 + \sqrt{3})^2 + 1^2 = 4 + 4\sqrt{3} + 3 + 1 = 8 + 4\sqrt{3}$$

22 (Poland). When it is given a list of four numbers, the Kangaroo Machine continues the list by typing the smallest non-negative integer that is different to each of the four preceding terms and then repeats this process over and over again. Jacob types in the numbers

2, 0, 2, 3,

into the machine. What number will be the 2023rd in the list?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

SOLUTION: The sequence of digits typed by the machine is: 2, 0, 2, 3, 1, 4, 0, 2, 3, 1, 4, 0, 2, 3, 1, 4, 0, 2, 3, ...

Starting from the second digit this sequence has a period of 5. Then a number in the 2023rd position is the same as a number in the 3rd position, i.e., 2.

23 (Australia). A rectangle with vertices $(0, 0)$, $(100, 0)$, $(100, 50)$ and $(0, 50)$ has a circle with centre $(75, 30)$ and radius 10 cut out of it. What is the gradient of the line through $(75, 30)$ that divides the remaining area of the rectangle into two equal parts?

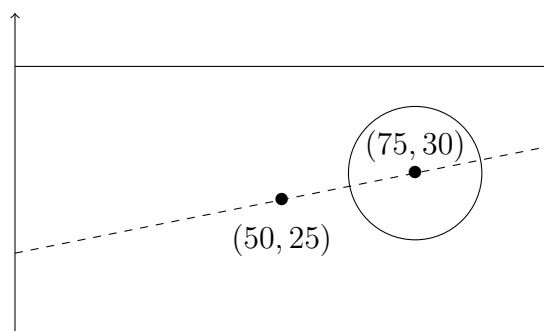
(A) $\frac{1}{5}$

(B) $\frac{1}{3}$

(C) $\frac{1}{2}$

(D) $\frac{2}{5}$

(E) $\frac{2}{3}$



SOLUTION: Any line through the centre of the circle $(75, 30)$ bisects the area of the circle. Any line through the centre of the rectangle $(50, 25)$ bisects the area of the rectangle. So, since the circle lies wholly within the rectangle, the line passing through the centre of the rectangle and the centre of the circle must cut the remaining area in half, as a semicircle is cut out of each side. So, the gradient of the line dividing the remaining area of the rectangle into two equal parts is the gradient of the line that passes through $(50, 25)$ and $(75, 30)$ and so has a gradient of $\frac{30 - 25}{75 - 50} = \frac{1}{5}$.

24 (Turkey). When Metin's phone is fully charged, it runs out in 32 hours if he only uses it for phone-calls, in 20 hours if he only uses it for the internet, and in 80 hours if he does not use it at all. Metin gets on a train with his phone half-charged. While on the train, the time he is on the internet, the time he is making phone-calls and the time he is not using it are all the same. His phone runs out of charge just as the train reaches his destination. How many hours did the train journey take?

(A) 10

(B) 12

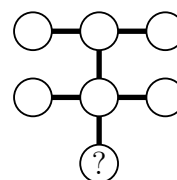
(C) 15

(D) 16

(E) 18

SOLUTION: Because Metin's phone is half charged, the times the battery will run out are 40, 16, and 10 hours respectively. Let say the journey took x hours. Then, it used $x/120$ and $x/48$ and $x/30$ from battery and their total is 1. We find $x = 16$ from here.

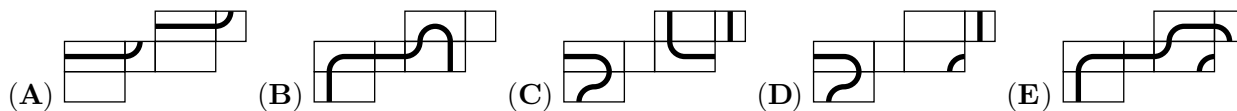
25 (Greece). Seven different single-digit numbers are written in the circles of the diagram shown with one number in each circle. The product of the three numbers in each of the three lines of three numbers is the same. Which number is written in the circle containing the question mark?



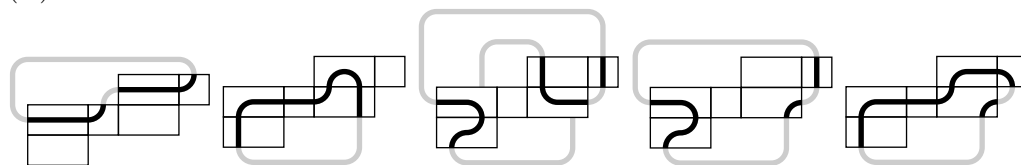
- (A) 2 (B) 3 (C) 4 (D) 6 (E) 8

SOLUTION: It is easy to see that neither 5 nor 7 can be used because each number is at most on two sets of three numbers, so the third set cannot have the same product as each of the other two. The product of all the numbers in the two horizontal rows must be a perfect square (the square of the common product). But the product of all seven digits that appear is $2 \cdot 3 \cdot 4 \cdot 6 \cdot 8 \cdot 9 = 2^7 \cdot 3^4$. So the number on the circle with the question mark can only be 2 or $2^3 = 8$. But it cannot be 8 because then the product of each row would be $\sqrt{2 \cdot 3 \cdot 4 \cdot 6 \cdot 9} = 36$, but this is not a multiple of 8. So the number in the circle is 2. If we want an example to see that this is possible, we have: First row 1, 9, 8, second row 3, 4, 6 and column 9, 4, 2, with common product 72.

26 (Slovenia). Leon has drawn a closed path on a cuboid and then unfolded it to give a net. Which of the nets shown could not be the net of Leon's cuboid?



SOLUTION: The nets given can be folded into solid uniquely. Following the path one can see that only (C) is not closed.



Connections in grey show which pieces are glued together (i.e. belong to the same path).

27 (Hungary). How many three-digit positive integers x are there, such that subtracting the sum of digits of x from x gives a three-digit number whose digits are all the same?

- (A) 1 (B) 2 (C) 3 (D) 20 (E) 30

SOLUTION: Let the three digit number be $\overline{abc}$. Then, the difference between the number and the sum of its digits is $100a + 10b + c - a - b - c = 99a + 9b$. This difference is divisible by 9. Ergo, the result can only be 333 or 666 (easy to see, that 999 is not possible.) The number abc can be 340, 341, 342, ..., 349 or 680, 681, 682, ..., 689. There are 20 different numbers with that property.

28 (Hungary). In how many different ways can the word **BANANA** be read from the following table by moving from one cell to another cell with which it shares an edge? Cells may be visited more than once.

B	A	N
A	N	A
N	A	N

- (A) 14 (B) 28 (C) 56 (D) 84 (E) other value

SOLUTION: Case 1: The first N is in the 1st row, 3rd column. If the 2nd N is in the same, then there are $2 \cdot 2 = 4$ possibilities. If the 2nd N is in the 2nd row, 2nd column, then there are $2 \cdot 4 = 8$ possibilities. If the 2nd N is in the 3rd row, 3rd column, then there are $1 \cdot 2 = 2$ possibilities. Altogether, there are $4 + 8 + 2 = 14$ possibilities.

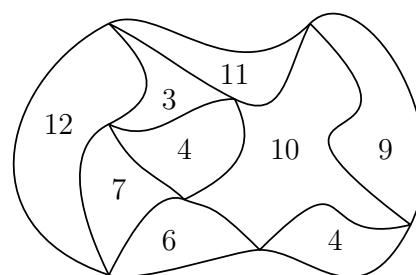
Case 2: The first N is in the 2nd row, 2nd column. If the 2nd N is in the same, then there are $4 \cdot 4 = 16$ possibilities. If the 2nd N is one of the other three, then there are $2 \cdot 2 = 4$ possibilities. Altogether, there are $2 \cdot (16 + 3 \cdot 4) = 56$ possibilities.

Case 3: The first N is in the 1st row, 3rd column. As in the first case, 14 possibilities.

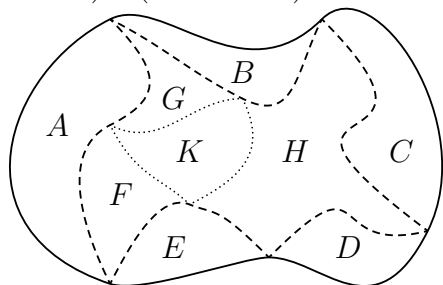
Answer: $14 + 56 + 14 = 84$.

29 (Greece). The diagram shows a map of a park. The park is divided into regions. The number inside each region gives its perimeter, in km. What is the outer perimeter of the park?

- (A) 22 km (B) 26 km
 (C) 28 km (D) 32 km
 (E) none of the previous



SOLUTION: The sum of the perimeters of A, B, C, D and E give the length of the outside line increased by the dashed line. If we subtract the perimeters of F, G and H , then we subtract the length of the dashed line but we have now also subtracted the length of the dotted line. So we add the length of the dotted line to compensate. In other words the perimeter of the park is $(A + B + C + D + E) - (F + G + H) + K$. Here it is $42 \text{ km} - 20 \text{ km} + 4 \text{ km} = 26 \text{ km}$.



30 (Mexico). Pia wants to write the integers 1 to 9 in the nine boxes shown so that the integers in any three adjacent boxes add to a multiple of 3. In how many ways can she do this?

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- (A) 6^4 (B) 6^3 (C) 2^9
 (D) $6 \times 5 \times 4 \times 3 \times 2 \times 1$ (E) $9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

SOLUTION: In order to have a sum divisible by 3 we need the remainders modulo 3 coming up in the following pattern $ABCABCABC$ with $A, B, C \in \{0, 1, 2\}$. There are $3 \cdot 2 \cdot 1 = 6$ possibilities to order the remainders 0, 1, 2 to the letters A, B, C . Once this is done there are 6 ways to arrange numbers 1, 4 or 7, 6 there are 6 ways to arrange numbers 2, 5 or 8 and there are 6 ways to arrange numbers 3, 6 or 9 to the places with the corresponding remainder. In total there are $6 \cdot 6 \cdot 6 \cdot 6 = 6^4$ possible desired distributions.